

AI/ML-Driven Glacial Lake Detection & Temporal Change Analysis

Autonomous Earth Observation & GLOF Early-Warning Intelligence

1. The Core Problem & Motivation

Due to global climate warming, glaciers in high-mountain ranges (Himalayas, Karakoram, Andes, Alps) are melting and retreating at unprecedented rates. As glaciers retreat, meltwater pools behind unstable natural dams made of loose moraine rock, sediment, and ice.

These bodies of water are called **Glacial Lakes**.

The Danger: Glacial Lake Outburst Floods (GLOFs)

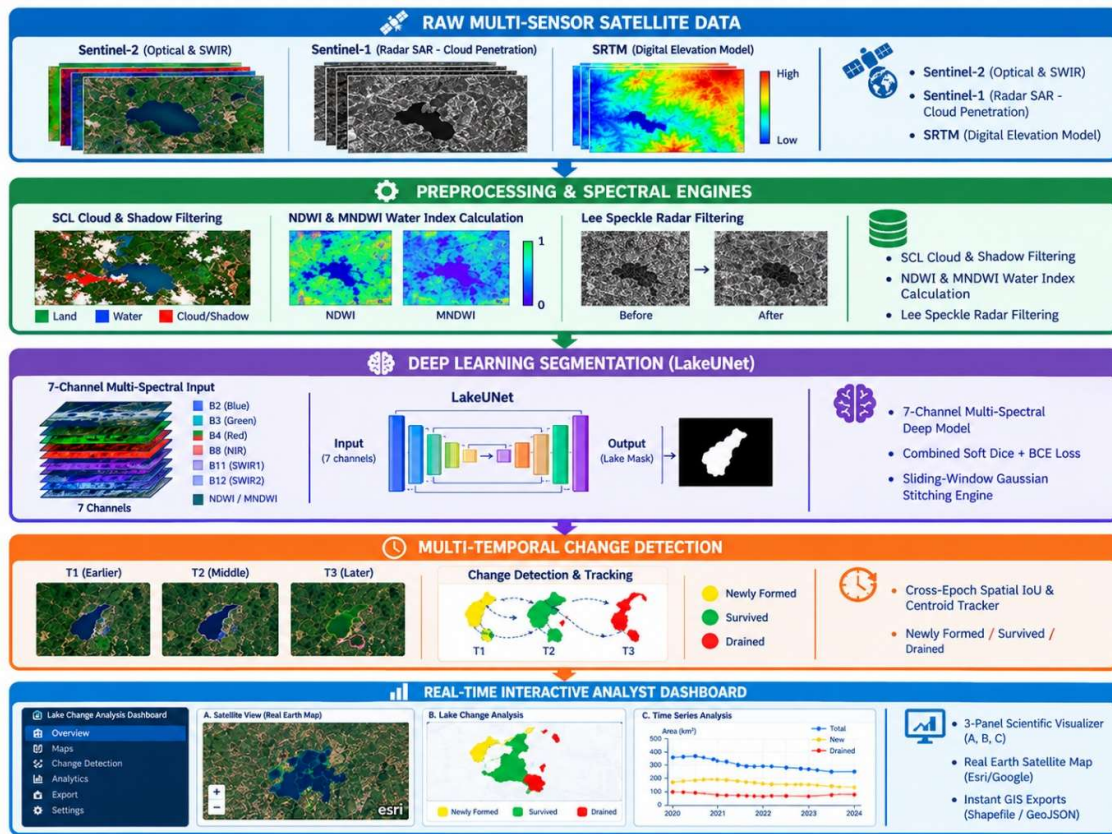
When an unstable natural dam breaches—triggered by an avalanche, earthquake, rockfall, or water pressure—millions of cubic meters of water and debris cascade downstream within minutes. This destructive phenomenon is known as a **GLOF**. It can wipe out downstream mountain villages, roads, bridges, and hydroelectric power stations.

The Challenge with Traditional Monitoring

- **Extreme Terrain:** Mountain valleys sit between 3,500 m and 6,500 m altitude; field surveys are dangerous, slow, and expensive.
- **Complex Satellite Signatures:** Dark mountain cast shadows, frozen lakes, seasonal snowpack, and clouds mimic or obscure water bodies on optical satellite imagery.
- **Scale:** Thousands of isolated glacial valleys need continuous, multi-year monitoring.

2. Our Solution: What Does This System Do?

This project is an **end-to-end automated AI/ML geospatial intelligence platform**. It takes multi-sensor satellite imagery over high-altitude mountain ranges, automatically detects and delineates glacial lakes, compares changes across different years (e.g., 2016 vs. 2022), categorizes their behavioral status, and flags potential hazards on an interactive GIS dashboard.



3. Step-by-Step Architecture & Pipeline Breakdown

Step 1: Multi-Sensor Data Ingestion & Engineering

Instead of relying on standard 3-color RGB imagery (which easily confuses dark mountain shadows with water), the pipeline fuses data across multiple spectral bands:

Sensor / Layer	Band / Index	Why It Is Crucial
Sentinel-2 Optical	Blue, Green, Red (B02, B03, B04)	True-color baseline and visual context.
Near-Infrared (NIR)	Band 08	Water strongly absorbs NIR, making lake boundaries contrast sharply against rocks.
Shortwave-IR (SWIR)	Band 11	Separates snow/ice from liquid water (prevents false alarms on glaciers).
Spectral Water Indices	NDWI & MNDWI	Formulated mathematical ratios that isolate liquid water signatures: $NDWI = \frac{Green - NIR}{Green + NIR}$

Sensor / Layer	Band / Index	Why It Is Crucial
Radar (Sentinel-1 SAR)	C-Band VV/VH Backscatter	Radar waves pierce through dense clouds; flat water acts as a mirror, producing near-zero backscatter.
Elevation (DEM)	SRTM 30 m Topography	Slope and aspect analysis filters out steep shadowed cliffs that mimic dark water.

Step 2: Deep Learning Segmentation Engine (LakeUNet)

- **Architecture:** An encoder-decoder **U-Net** neural network with high-resolution skip connections. The encoder extracts high-level semantic features, and the decoder reconstructs precise pixel boundaries down to individual glacier margins.
- **Dual-Mode Capability:**
 1. **Primary Multi-Spectral Model (7 Channels):** Ingests [B2, B3, B4, B8, B11, NDWI, DEM].
 2. **Fast RGB Fallback Model (3 Channels):** Allows users to upload standard satellite captures or drone screenshots (JPG/PNG) without multi-spectral bands.
- **Tackling Class Imbalance:** High-mountain lakes occupy less than 2% of the total landscape. Standard cross-entropy loss would bias the network toward predicting "no lake". We deploy a hybrid loss: $\mathcal{L}_{\text{total}} = 0.5 \cdot \mathcal{L}_{\text{Dice}} + 0.5 \cdot \mathcal{L}_{\text{BCE}}$
- **Sliding-Window Inference with Gaussian Blending:** The model processes large scenes in 256×256 tiles and stitches them using Gaussian edge weighting, eliminating visible seamline artifacts.

Step 3: Vectorization & Boundary Refinement

1. The raw binary probability mask undergoes morphological opening/closing to eliminate stray pixel noise.
2. The raster mask is vectorized into geospatial polygons using **Rasterio** and **GeoPandas**.
3. **Douglas-Peucker simplification** smooths pixelated jagged staircase boundaries into smooth shorelines while preserving true surface area.

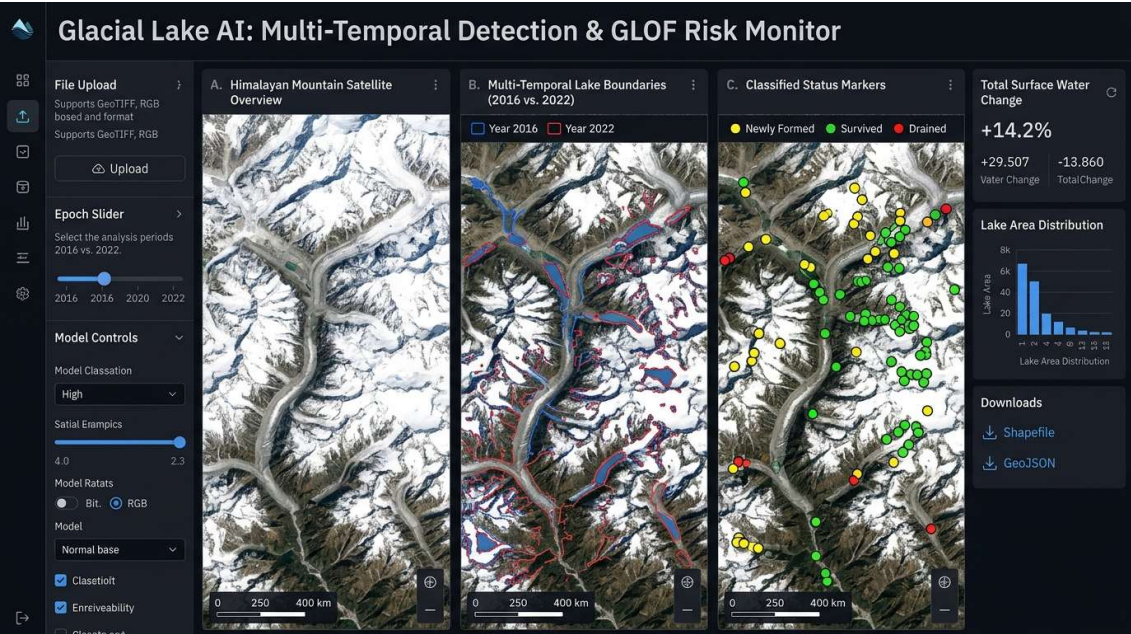
Step 4: Multi-Temporal Change & Risk Classification

The system takes lake polygons from an earlier baseline epoch (e.g., Year 2016) and a later comparison epoch (e.g., Year 2022). Using **Intersection-over-Union (IoU)** overlap and centroid proximity, each lake is assigned a classification status:

Classification	Visual Marker	Identification Rule	GLOF Risk Meaning
 Newly Formed	Yellow Pin	Exists in 2022, absent in 2016.	High Priority: Young lakes dammed by fresh, unconsolidated loose moraines prone to sudden initial failure.
 Survived	Green Pin	Exists in both 2016 and 2022.	Monitoring: Tracks area growth ($\Delta\text{Area}\%$). Lakes expanding rapidly ($>20\%$) are flagged for alert.
 Drained	Red Pin	Present in 2016, gone in 2022.	Investigate: Indicates a past outburst flood, underground subglacial drainage, or complete dry-up.

 4. The Interactive User Experience (Dashboard)

The user interacts with the system through a web dashboard built with Streamlit and Folium:



- 1. **Top Header:** Project identity and real-time status indicators.
- 2. **Left Sidebar:** Upload your own satellite scenes (GeoTIFF, JPG, PNG), select observation years, and fine-tune sensitivity thresholds and cloud/shadow filters.
- 3. **Center 3-Panel Scientific Visualizer:**

- **Panel A:** High-altitude satellite overview of the glacier basin.
 - **Panel B:** Multi-temporal boundary outlines (Year 2016 in **Blue** vs. Year 2022 in **Red**).
 - **Panel C:** Classified markers ( **Newly Formed**,  **Survived**,  **Drained**).
4. **Live Real Earth Satellite Map:** An interactive Folium/Leaflet viewer powered by high-resolution **Esri World Imagery** and **Google Earth Satellite** basemaps, complete with live coordinates, distance measurement tools, and clickable lake attribute popups.
5. **Right Analytics & Direct GIS Downloads:**
- Instant KPIs: Net water surface expansion (+14.2%), new lake counts, and drained lake counts.
 - Dynamic lake area distribution bar chart.
 - One-click downloads for **ESRI Shapefile ZIP**, **GeoJSON**, and **CSV summary logs** ready to open directly in QGIS, ArcGIS, or Google Earth.

5. Real-World Impact & Applications

-  **Early-Warning for GLOF Hazards:** Automatically flags rapidly expanding lakes sitting above mountain villages and hydropower infrastructure.
-  **Autonomous Remote Sensing:** Eliminates tedious manual on-screen digitizing by human analysts across thousands of square kilometers of mountain terrain.
-  **GIS-Ready Outputs:** Direct integration into national disaster management databases (NDMA, ICIMOD, GLIMS) via standard vector shapefiles.
-  **Anywhere on Earth:** Adaptable to any glacial region across the globe, from the Karakoram and Himalayas to the Andes and Arctic.